

Phil 7:

Some notes on Assignment 3

General points:

1. The anti-plagiarism **challenge** continues!
2. Sequence of **assignments**:
 - **A1**: description + logic.
 - **A2**: description + logic + evaluation.
 - **A3**: description + logic + evaluation + argument.

 - Use previous assignments as models.
3. Feedback from A2:
 - **Overall significant improvements**: clarity, accuracy, general logic.
 - **But**: claims without reasons.
 - Unorganized reasons.
 - Stating an argument v. disconnected reasoning.

Assignment 3

Due Date

11:59 PM, Sunday, November 17, 2019

Topic & Structure

1000 words.

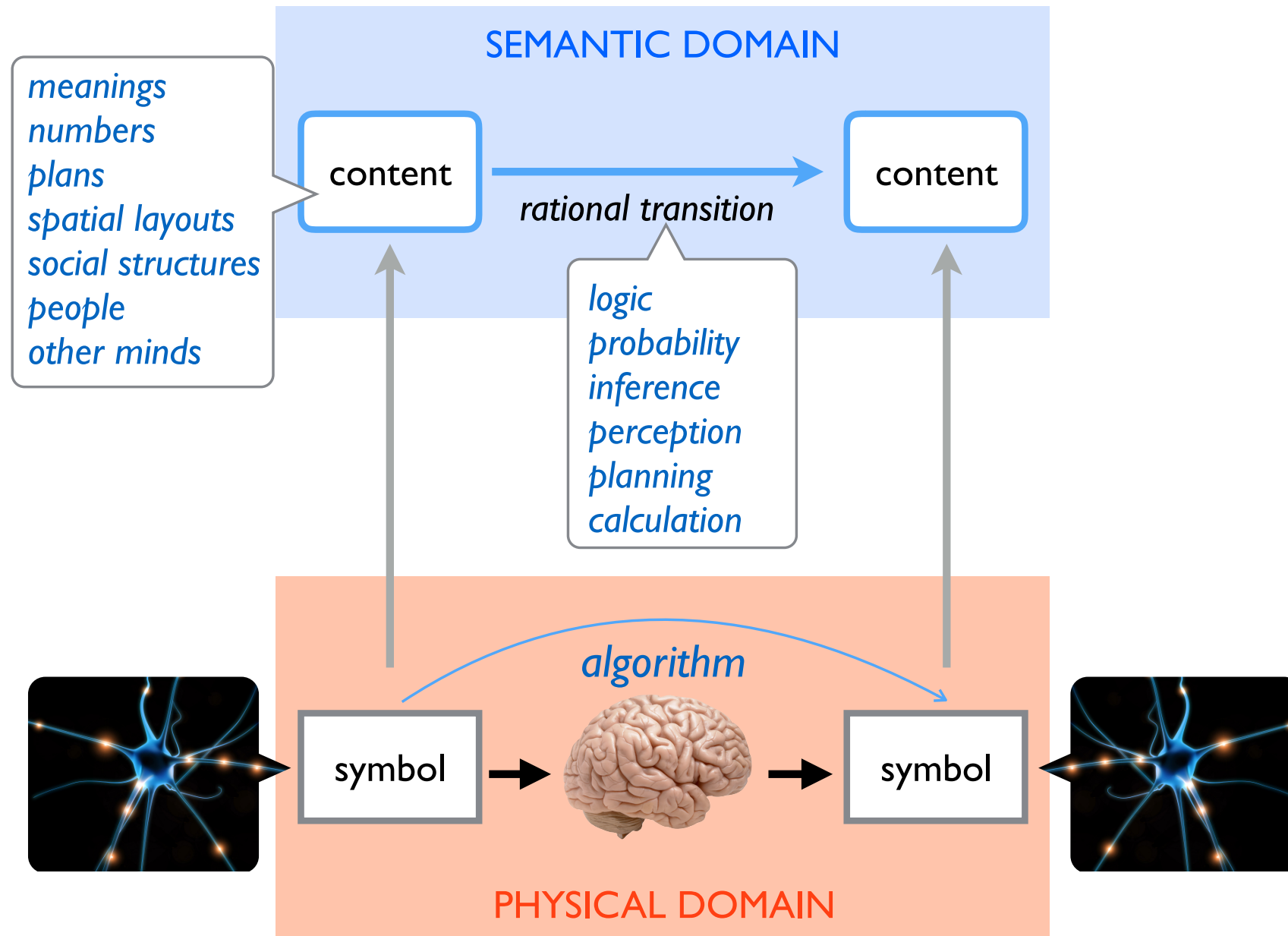
Evaluate the Computational Theory of Mind.

Assignment 3

1. **Summarize U2: Computationalism** in one paper.
2. **Flow of Argument**
 - Part I: Computationalism
 - Part II: The Chinese Room Argument
 - Part III: Evaluating the ArgumentTry to make each section relate to the last.

Part I: Computationalism: 2 paragraphs

- (1) **Explain the idea of computationalism.** Address the roles played by symbols, computers, mental representations, and cognition.
- (2) **Illustrate your discussion with an example** (actual or hypothetical) that shows how computationalism is supposed to work.

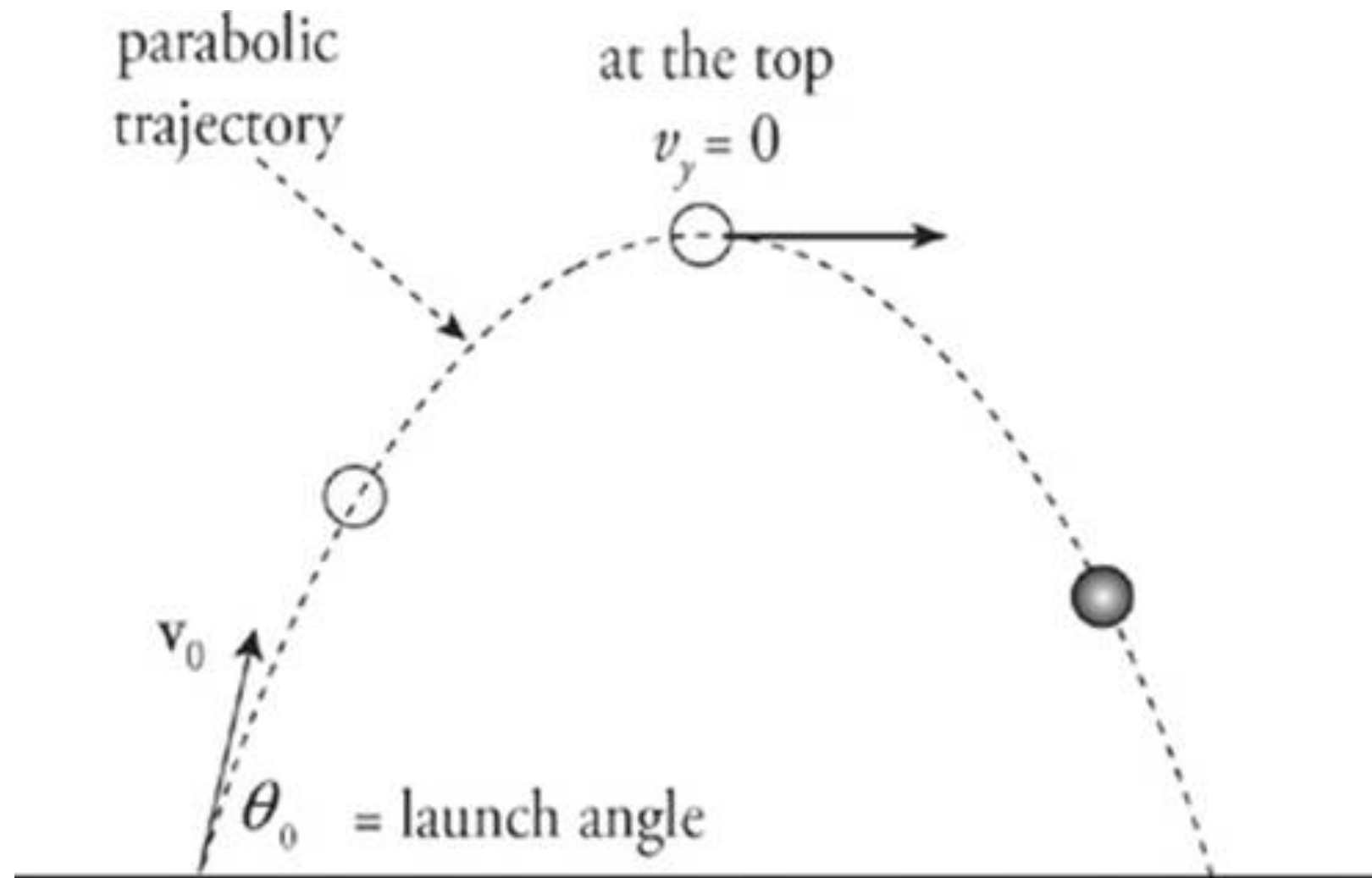


Note about examples of computationalism:

You can use either an everyday example (e.g. estimating the trajectory of a ball in the air) or a more scientific example (e.g. path integration in desert ants).

You can use an example discussed in class or one of your own design. Your example can be merely hypothetical, but it is good to be specific. Some suggestions:

- estimating the trajectory of a ball in the air
- path integration among desert ants
- calculating a simple mathematical function (like times 2, or binary +1)
- logical reasoning (if/then reasoning)
- decision making (means/end reasoning)
- navigation
- memory
- visual perception
- blind sight
- implicit bias



(current location, destination) $\longrightarrow$ (force, angle, direction)

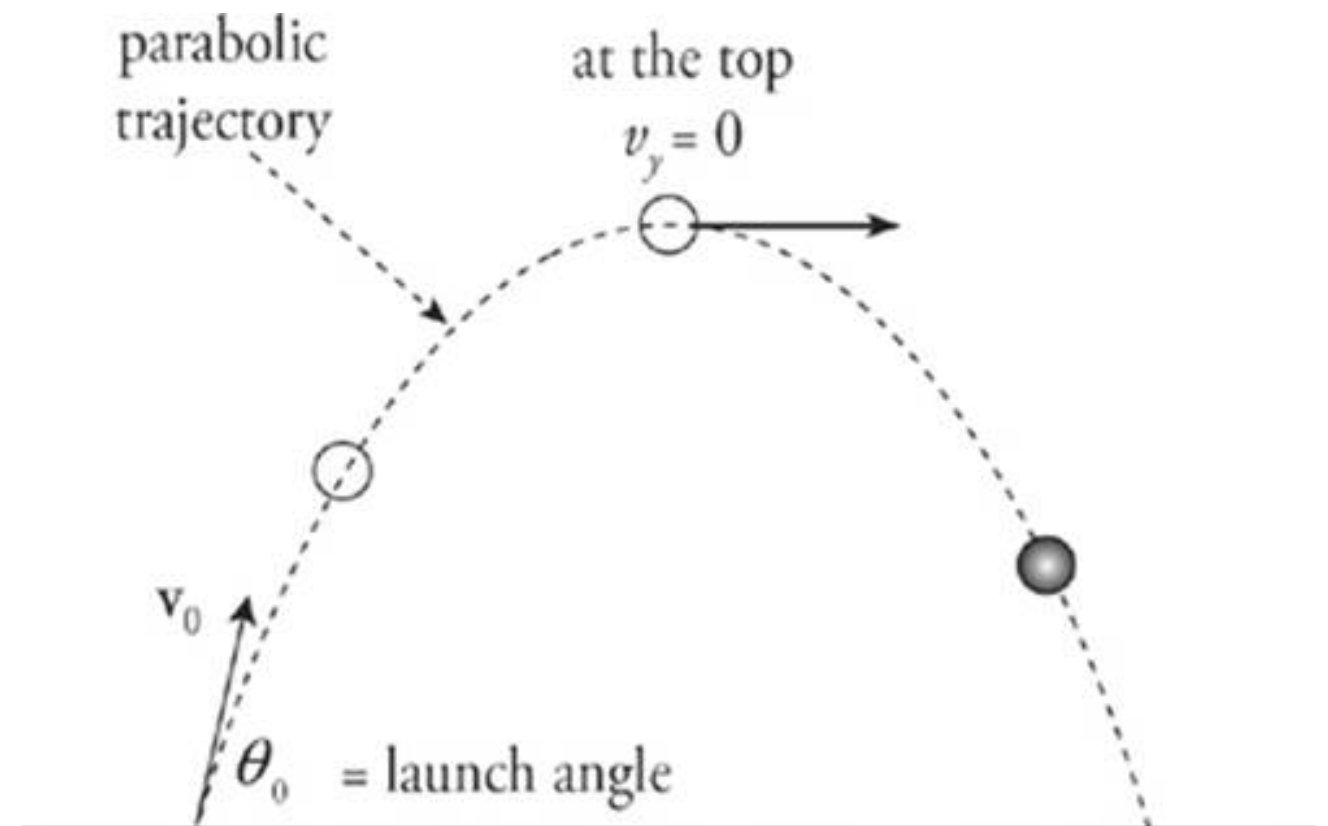
INPUT

OUTPUT

(current location, destination)



(force, angle, direction)



Part II: The Chinese Room Argument: 3 paragraphs

- (1) **Introduce the Chinese Room Argument:** what is the set up, and the key facts?
- (2) **Explain the point** of the Chinese Room Argument, highlighting the two key premises. What does each say? Why is each plausible?
- (3) **Explain the conclusion** of the Chinese Room Argument. What does this conclusion tell us about computers and the mind? How is the argument related to the problem of meaning?



The Chinese Room Argument

- ▶ 1. If computationalism is true,
then T'Challa must understand Chinese.
- 2. T'Challa doesn't understand Chinese.
- 3. Therefore: computationalism is false.

The Systems Reply

“Some people attempt to answer the Chinese room example by saying that... the whole system, including the room, the baskets full of symbols and the ledgers containing the programs and perhaps other items as well, taken as a totality, that understands Chinese.”

The Robot Reply

“Another common response is to imagine that we put the Chinese understanding program inside a robot. If the robot moved around and interacted causally with the world, wouldn't that be enough to guarantee that it understood Chinese?”

Part III: Evaluation: 2 paragraphs

- (1) **Explain either the Systems Reply or the Robot Reply.** Suggestions: explain both the negative and positive claims involved in this reply. How is the reply supposed to undermine the argument? What does it say about the problem of meaning instead?
- (2) **Evaluate: in light of your discussion, is the Chinese Room Argument successful** (i.e. does it prove that computationalism is false)? If you think the argument works, how do you respond to the reply discussed above? If you think the reply discussed above defeats the argument, what reason do you have for believing that reply?



<Hi! I'm Zoey. I'm a
mad scientist. And a
roller derby
champion! >

Key points

1. **Robot Reply:** does this necessarily involve systems reply?
Yes and no. It necessarily involves the idea of *emergence*, but differs about the underlying physical basis of the mind.
2. **Implicit bias and CRA:** The Systems Reply and Robot Reply both claim that cognition emerges from non-mental, physical basis. Searle might object that *consciousness* is the root of meaning, so the Systems/Robot Reply won't work. Implicit bias undermines this objection. It suggests you can have cognition (and meaning) without consciousness.
3. **Which version of CTM?** The one you think is most plausible and comprehensible.

Key points

1. Suggested readings:

- **CTM:** Pinker “Standard equipment,” Goldman “Desert Ants,” Clark “Meat Machines,” Ravenscroft “Computational Theory of Mind”
- **CRA:** Searle “Can Computers Think?”
- **Replies:** Systems reply reading, externalism reading, implicit bias reading.

What is **meaning**?

- It's a technical term which describes mental representations.
- It **doesn't** have anything to do with *importance, significant personal experiences, the meaning of life, or implications for culture and history.*
- Meaning is *about-ness*. Are your mental states **about** anything?
- Examples:
 - The sentence "The earth is flat" is **about the earth**.
 - My belief that it is snowing in Chicago is **about snow**.
 - Your desire to own a G-Class is **about that model of car**.
 - My perception that there is a desk in front of me is **about the desk**.
- How could a purely physical state be *about* anything? This is the **problem of meaning**.